Jittering Effects Analysis and Beam Training Design for UAV Millimeter Wave Communications

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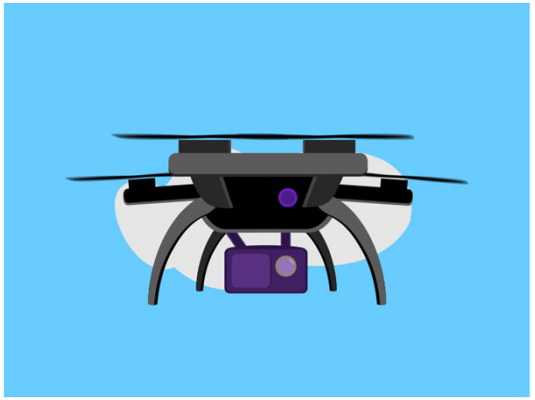
1 Recent Work

[1] **W. Wang** and W. Zhang, "Jittering effects analysis and beam training design for UAV millimeter wave communications," *IEEE Transactions on Wireless Communications.* DOI: 10.1109/TWC.2021.3118558

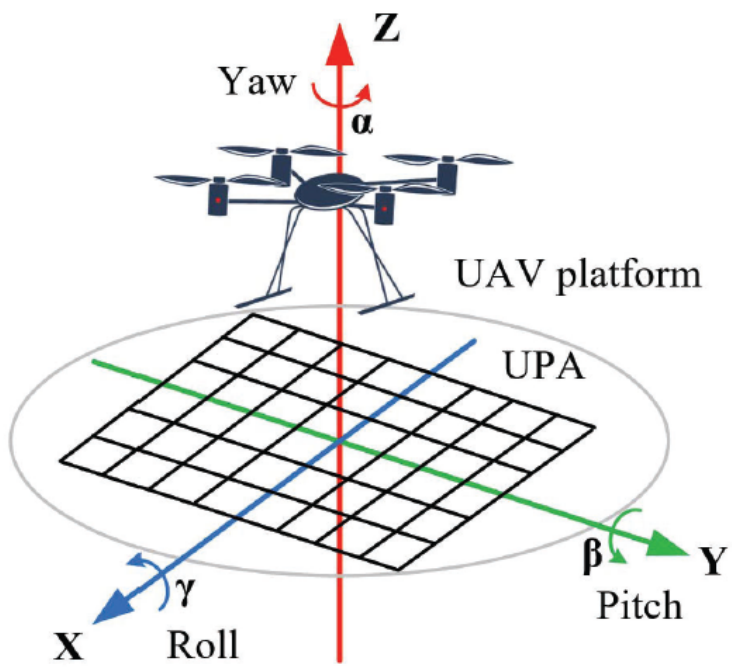
Research Aim:

(1) Understand the effects of UAV jitter to UAV mmWave channel

(2) Design a beam training scheme for UAV mmWave communications



2.1 What is UAV jitter?



- UAV jitter: the unintended high-frequency change of UAV attitude/orientation

- Euler angles describe the orientation of a rigid body with respect to a fixed coordinate system

$$\mathbf{R} = \mathbf{R}_{Yaw}(\alpha)\mathbf{R}_{Pitch}(\beta)\mathbf{R}_{Roll}(\gamma)$$

- UAV jitter modelled by Euler angles

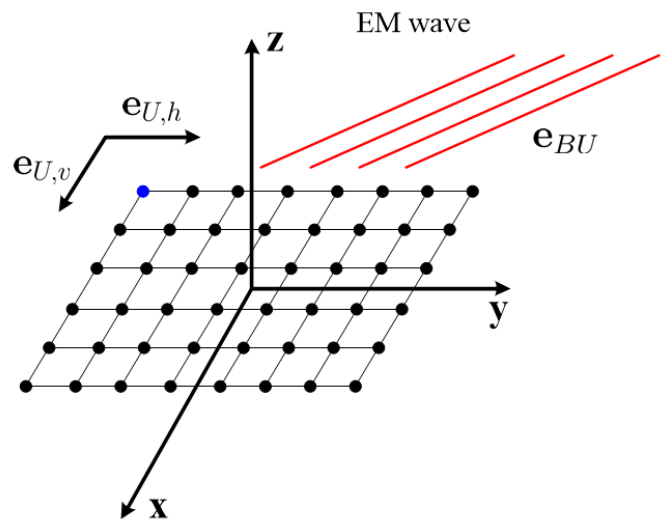
$$\alpha = \bar{\alpha} + \Delta\alpha$$
$$\beta = \bar{\beta} + \Delta\beta$$
$$\gamma = \bar{\gamma} + \Delta\gamma$$

Desired attitude angles: $\bar{\alpha}, \bar{\beta}, \bar{\gamma}$
Fluctuations caused by UAV jitter: $\Delta\alpha, \Delta\beta, \Delta\gamma$

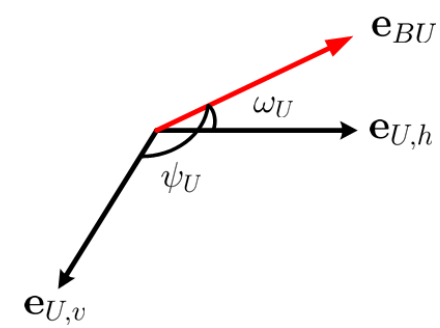
2.2 Narrow-band mmWave Channel Model

- Channel model:

$$\mathbf{H} = \sum_{l=0}^{L-1} \beta_l \mathbf{v}(\Psi_{U,l}, \Omega_{U,l}) \mathbf{v}^H(\Psi_{B,l}, \Omega_{B,l})$$



- $\mathbf{v}(\Psi_U, \Omega_U)$ and $\mathbf{v}(\Psi_B, \Omega_B)$ are array response (steering) vectors at UAV side and BS side.
- LoS component dominates, i.e., $|\beta_0| >> |\beta_l|, l \neq 0$
- **H** is characterized by $(\Psi_{U,0}, \Omega_{U,0})$ and $(\Psi_{B,0}, \Omega_{B,0})$ which are termed as **cosine AoA/AoD** or **direction cosines.**

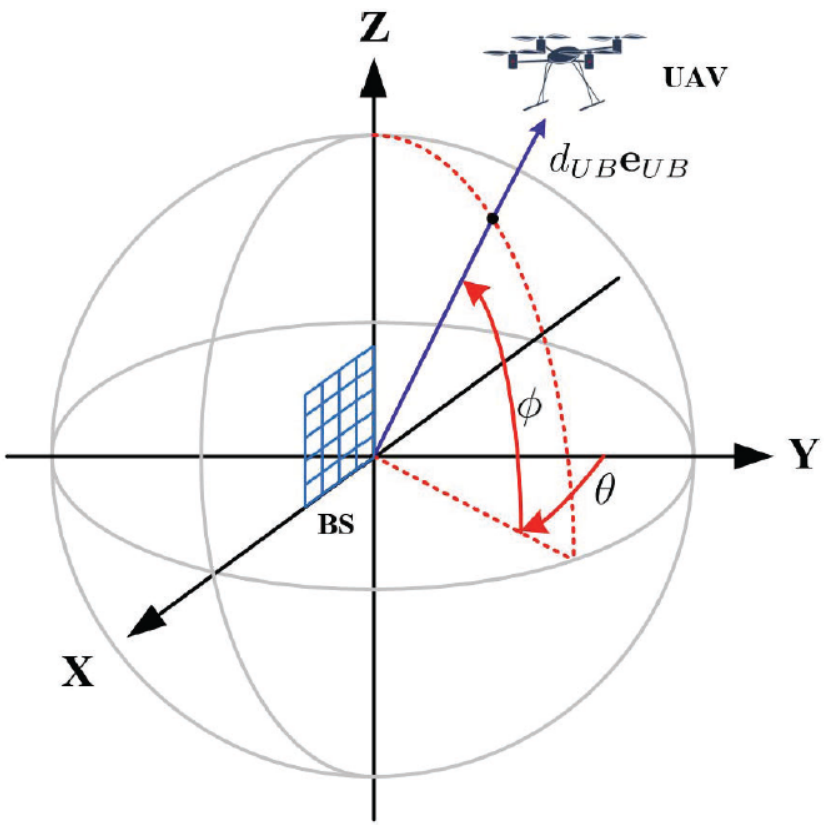


$$\Omega_U = \cos\omega_U = \mathbf{e}_{U,h}^T \mathbf{e}_{BU}$$

$$\Psi_U = \cos\psi_U = \mathbf{e}_{U,v}^T \mathbf{e}_{BU}$$

Jittering effects to UAV mmWave channel are primarily reflected in $(\Psi_{U,0}, \Omega_{U,0})$ $(\Psi_{B,0}, \Omega_{B,0})$

2.3 A Deep Look Into Angles



(1) Azimuth angle & Elevation angle

- Angles in spherical coordinate that are used to identify the position of UAV
- The direction vector from UAV to BS is

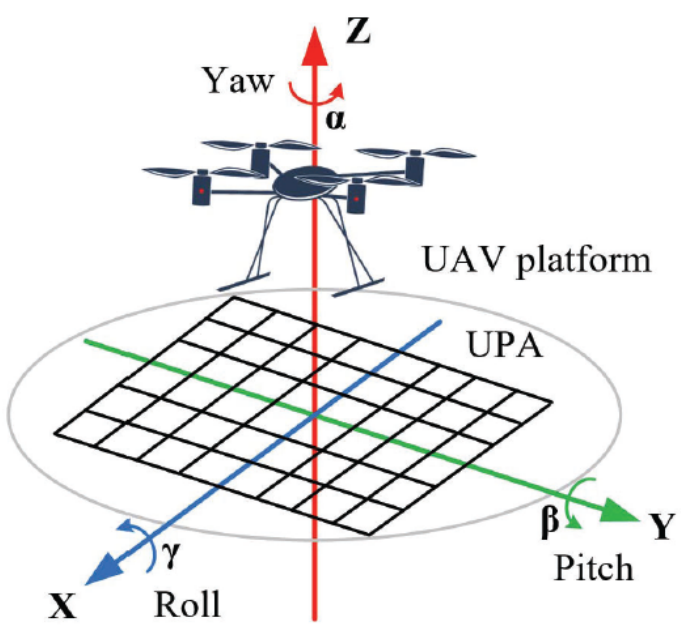
$$\mathbf{e}_{BU} = [\cos\phi\cos\theta,\ \cos\phi\sin\theta,\ \sin\phi]^T$$

where ϕ is elevation angle and θ is azimuth angle of the LoS path.

- Can be obtained via GPS and barometer.

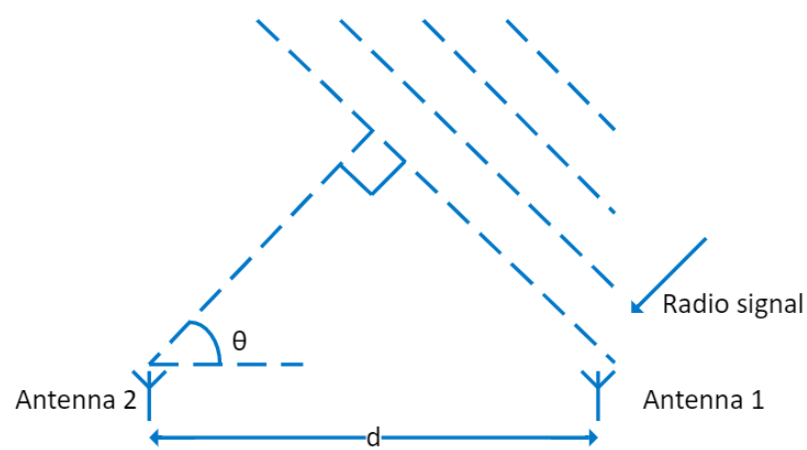
2.3 A Deep Look Into Angles

(2) Yaw, pitch & roll angles



- Can be measured via Gyroscope.

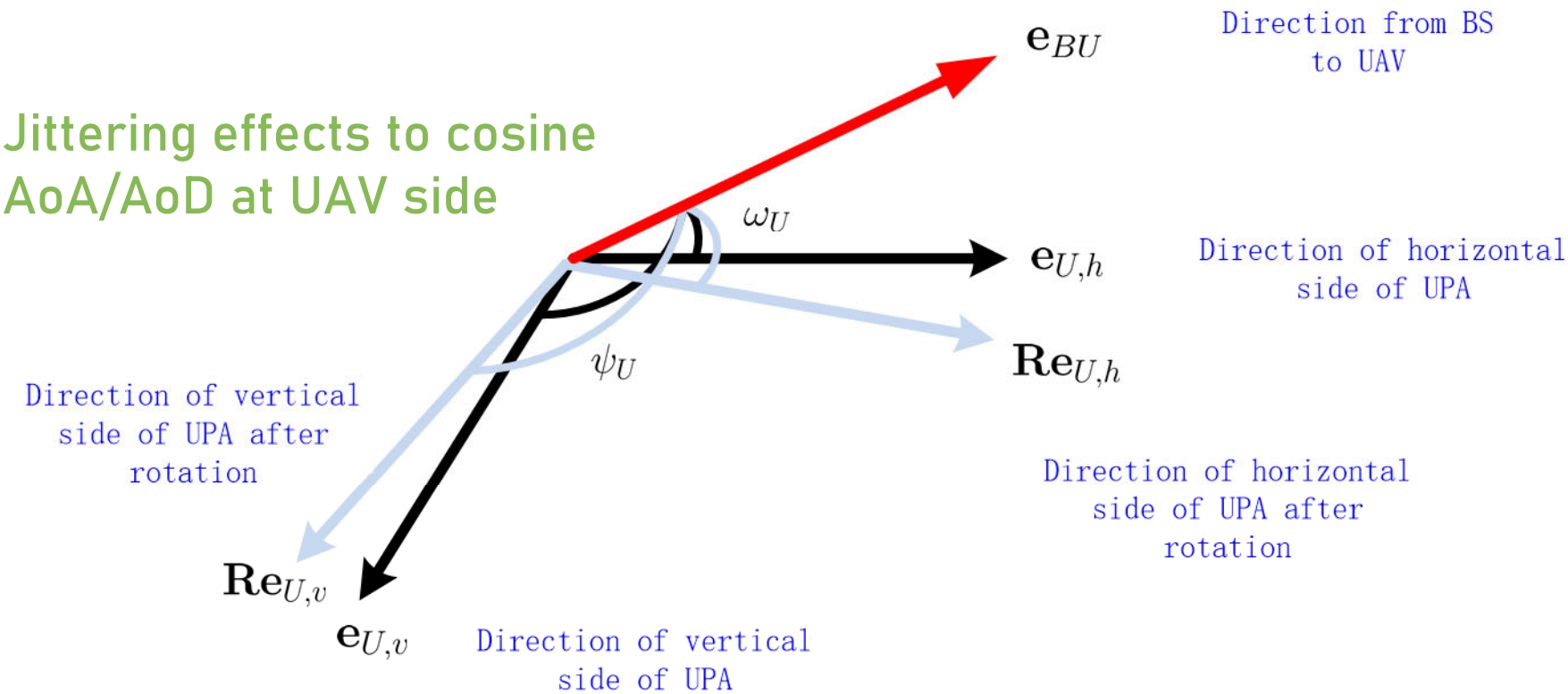
(3) Angle of arrival (AoA) & Angle of departure (AoD)



- Can be obtained through estimating phase differences of the elements of an antenna array.

2.3 A Deep Look Into Angles

Jittering effects to cosine AoA/AoD at UAV side



$$\Omega_U = \cos\omega_U = \mathbf{e}_{U,h}^T \mathbf{e}_{BU}$$

$$\Psi_U = \cos\psi_U = \mathbf{e}_{U,v}^T \mathbf{e}_{BU}$$

Rotation: R

$$\Omega_U = \cos\omega_U = \mathbf{e}_{U,h}^T \mathbf{R}^T \mathbf{e}_{BU}$$

$$\Psi_U = \cos\psi_U = \mathbf{e}_{U,v}^T \mathbf{R}^T \mathbf{e}_{BU}$$

2.4 Jittering Effects to Cosine AoA/AoD

Two-dimensional AoA/AoD	UAV position related angles (i.e., ϕ, θ)	UAV attitude related angles (i.e., α, β, γ)
(Ψ_B, Ω_B)	Dependent	Independent
(Ψ_U, Ω_U)	Dependent	Dependent

Table. Dependency of the cosine AoA/AoD (BS side and UAV side) on UAV position and UAV attitude

- **UAV Jittering effects to cosine AoA/AoD at UAV side cannot be ignored**
- **UAV Jittering effects to cosine AoA/AoD at BS side can be ignored**

2.4 Jittering Effects to Cosine AoA/AoD

Modelling of UAV jitter: $\Delta\alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \Delta\beta \sim \mathcal{N}(0, \sigma_\beta^2), \Delta\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$
$\Delta\alpha, \Delta\beta, \Delta\gamma$ are independent of each other.

Theorem 2. The two dimensional AoA/AoD at UAV side, i.e., $[\Psi_U(\alpha,\beta,\gamma), \Omega_U(\alpha,\beta,\gamma)]^T$, follows a Gaussian distribution, i.e., $[\Psi_U(\alpha,\beta,\gamma), \Omega_U(\alpha,\beta,\gamma)]^T \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, whose mean vector is

$$\boldsymbol{\mu} = \begin{bmatrix} \Psi_U(\bar{\alpha}, \bar{\beta}, \bar{\gamma}) \\ \Omega_U(\bar{\alpha}, \bar{\beta}, \bar{\gamma}) \end{bmatrix} \tag{32}$$

and covariance matrix is

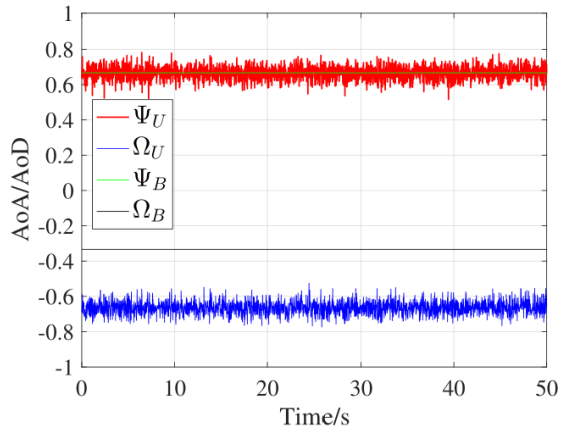
$$\boldsymbol{\Sigma} = \mathbf{J}(\bar{\alpha}, \bar{\beta}, \bar{\gamma}) \begin{bmatrix} \sigma_\alpha^2 & 0 & 0 \\ 0 & \sigma_\beta^2 & 0 \\ 0 & 0 & \sigma_\gamma^2 \end{bmatrix} \mathbf{J}(\bar{\alpha}, \bar{\beta}, \bar{\gamma})^T$$

the Jacobian matrix $\mathbf{J}(\alpha,\beta,\gamma)$ is represented as

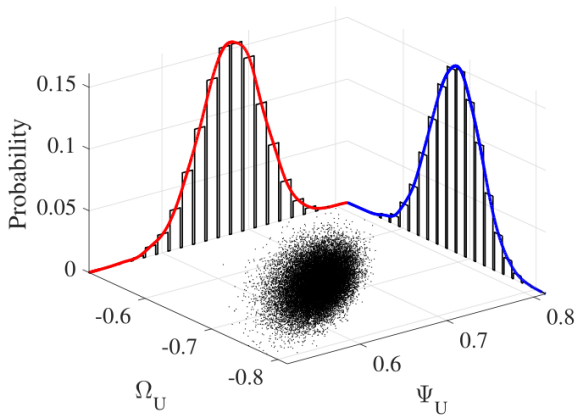
$$\mathbf{J}(\alpha,\beta,\gamma) = \begin{bmatrix} \frac{\partial \Psi_U}{\partial \alpha} & \frac{\partial \Psi_U}{\partial \beta} & \frac{\partial \Psi_U}{\partial \gamma} \\ \frac{\partial \Omega_U}{\partial \alpha} & \frac{\partial \Omega_U}{\partial \beta} & \frac{\partial \Omega_U}{\partial \gamma} \end{bmatrix}$$

2.4 Jittering Effects to Cosine AoA/AoD – Numerical Examples

Modelling of UAV jitter: $\sigma_\alpha = \sigma_\beta = \sigma_\gamma = 0.05$

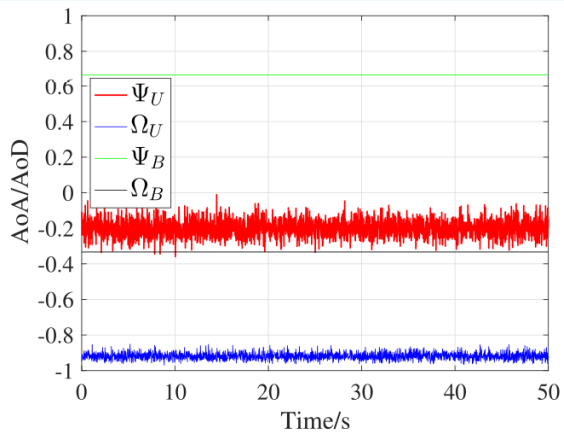


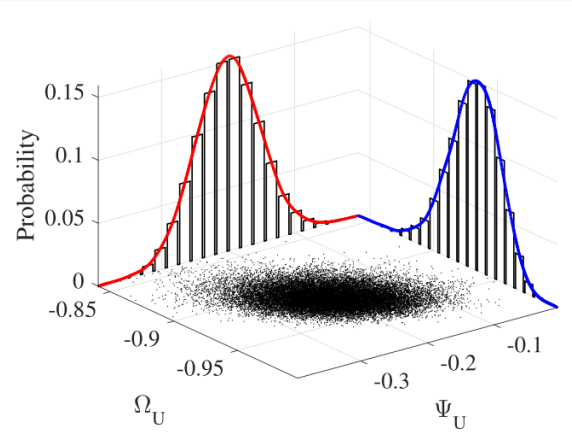
(a) Variation of Ψ_U, Ω_U, Ψ_B and Ω_B over time (Scenario 1)



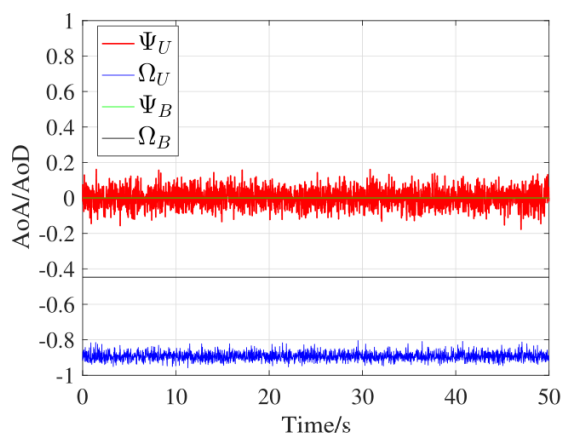
(b) Empirical marginal probability density function of Ψ_U and Ω_U (Scenario 1)

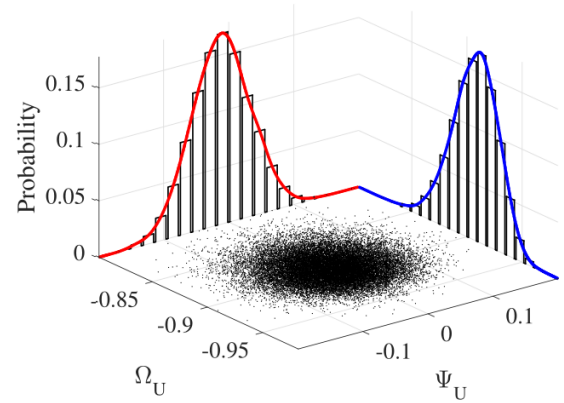
Scenario1: UAV is hovering at the position $\mathbf{p}_U = [-100, 100, 50]^T$ **(Cartesian coordinates) with its desired flight attitude being** $\bar{\alpha} = \bar{\beta} = \bar{\gamma} = 0$





Scenario2: UAV is hovering at the position $\mathbf{p}_U = [-100, 100, 50]^T$ **with its desired flight attitude being** $\bar{\alpha} = 1$ **rad, and** $\bar{\beta} = \bar{\gamma} = 0$ **rad**





(e) Variation of Ψ_U, Ω_U, Ψ_B and Ω_B over time (Scenario 3)

(f) Empirical marginal probability density function of Ψ_U and Ω_U (Scenario 3)

Scenario3: UAV is hovering at the position $\mathbf{p}_U = [0, 100, 50]^T$ **with its desired flight attitude being** $\bar{\alpha} = \bar{\beta} = \bar{\gamma} = 0$ **rad**

2.4 Jittering Effects to Cosine AoA/AoD

The same scale of of UAV jitter

+

A different UAV position

+

A different desired UAV attitude

Different scales of jittering effects to UAV mmWave channel

3.1 UAV Beam Training Under Jittering Effects

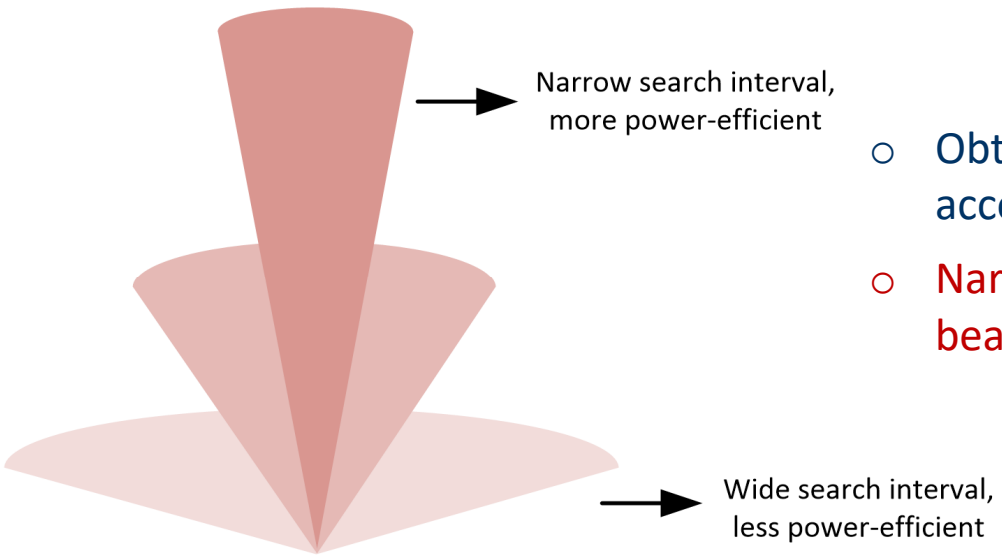
The Objective of Beam Training: Estimate AoA/AoD of the strongest path in mmWave channel to support the subsequent beam alignment operation.

- Ideally, with the relative position (Azimuth angle & Elevation angle) and attitude (Euler angles) of UAV, AoA/AoD can be accurately obtained.

- However, UAV platform faces the following challenges
  - Gyroscope and accelerometer are very sensitive to the jitter/vibration induced by the engine and wind gust;
  - The estimation error increases with the degree of UAV jitter.

3.1 UAV Beam Training Under Jittering Effects

- Utilizing navigation information to facilitate compressed sensing (CS) based UAV beam training.



- Obtain a rough estimate of cosine AoA/AoD according to navigation information.
- Narrow down the search interval of CS-based beam training

3.1 UAV Beam Training Under Jittering Effects

How to design the (random) CS sensing matrix within an arbitrary sensing range?

- Sensing matrix needs to satisfy restricted isometry property (RIP)
- Randomly generated sensing matrix satisfies RIP with high probability
  - Challenging, due to
    - (1) constant modulus of analog array antenna
    - (2) restrained sensing range

3.2 Construction of Direction-Constrained Sensing Matrix

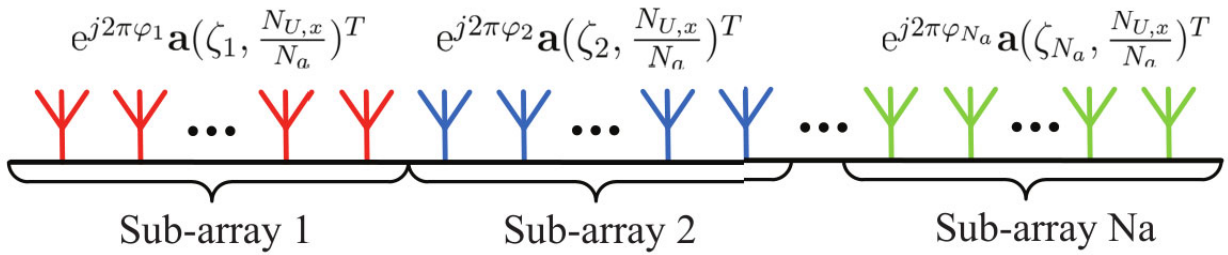


Fig. 7. Visualization of sub-array partition along one dimension

Sub-array based method:

Radiation range of the sub-array: $\left(-\frac{N_a}{N_{U,x}} + \zeta_{n_a}, \frac{N_a}{N_{U,x}} + \zeta_{n_a}\right)$

Step1. Restrict center angle ζ_{n_a} and sub-array size $N_{U,x}$ to restrain radiation range;

Step2. Randomize φ_{n_a} and partially randomize center angle ζ_{n_a} to satisfy RIP.

3.2 Construction of Direction-Constrained Sensing Matrix

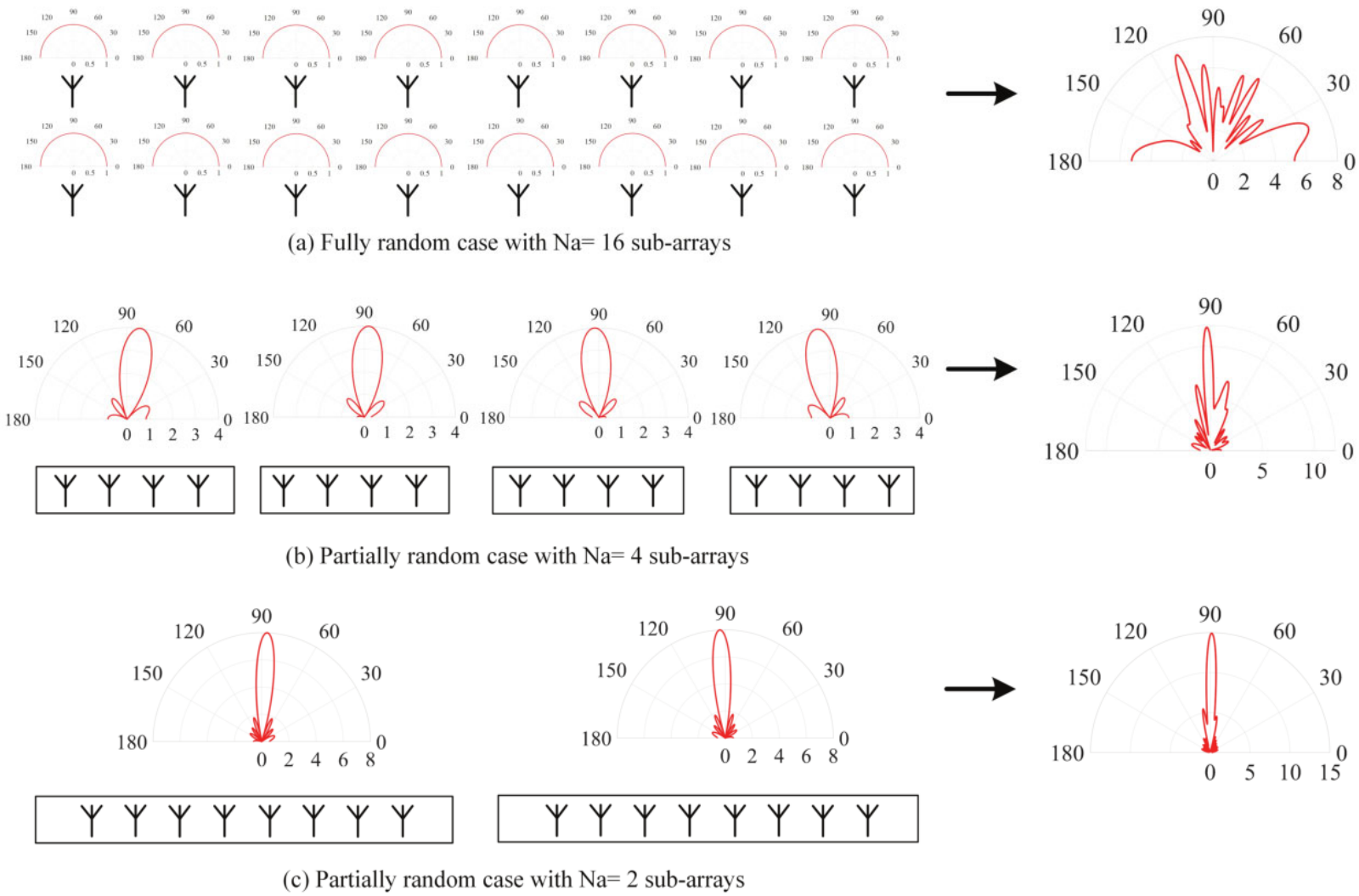
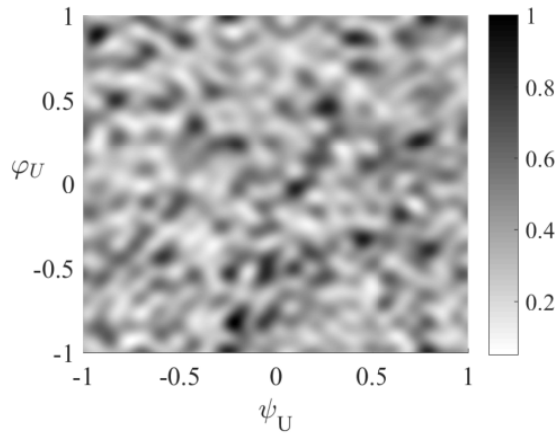
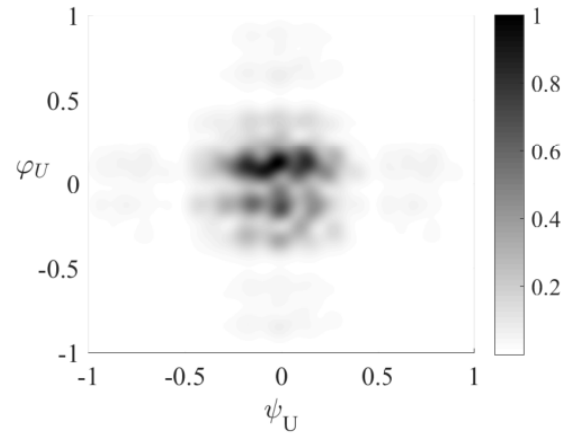


Fig. 8. Visualization of sub-array based design of random sensing vector (the scale of angle is $\cos^{-1}\Psi_U$)

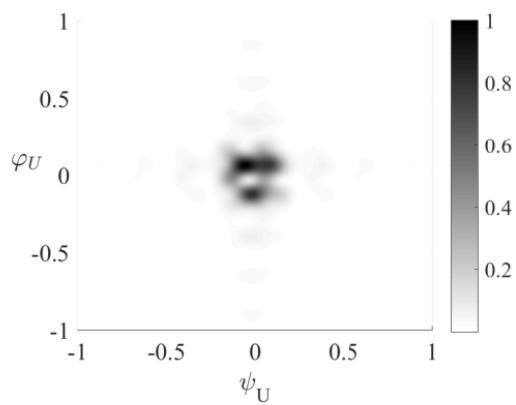
3.2 Construction of Direction-Constrained Sensing Matrix



(a) Fully random case with $N_a = 16$ sub-arrays, and the sensing range is $\Psi_U, \Omega_U \in (-1, 1)$



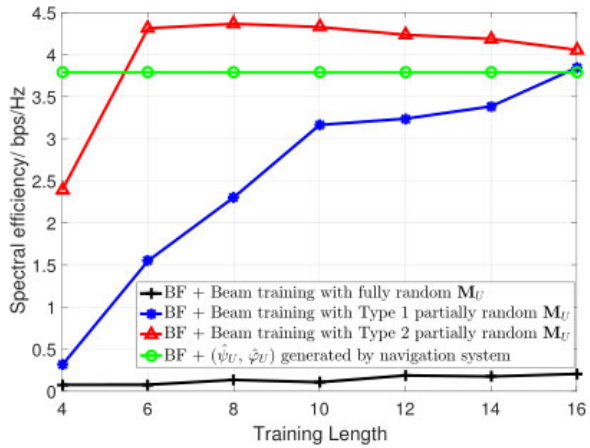
(b) Partially random case with $N_a = 4$ sub-arrays, and the sensing range is $\Psi_U, \Omega_U \in (-0.4, 0.4)$



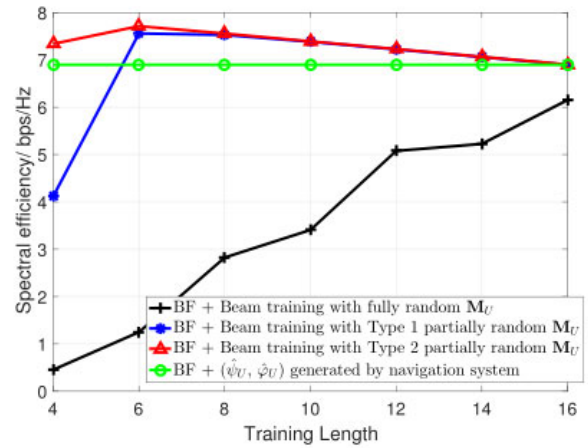
(c) Partially random case with $N_a = 2$ sub-arrays, and the sensing range is $\Psi_U, \Omega_U \in (-0.225, 0.225)$

Fig. 9. Beam space of fully and partially random sensing matrices

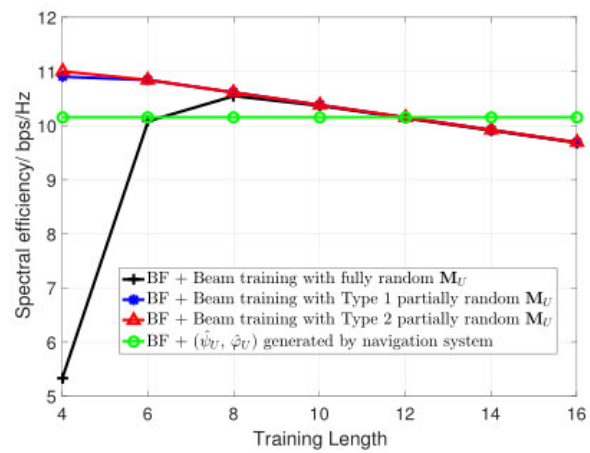
4 Numerical Results



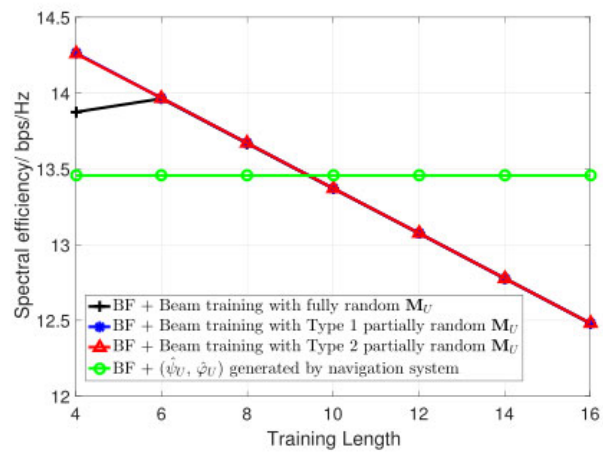
(a) When transmit power is -10dBm



(b) When transmit power is 0dBm



(c) When transmit power is 10dBm



(d) When transmit power is 20dBm

Fig. 12. Spectral efficiency of UAV beamforming enabled by the proposed beam training methods

5 Conclusion

- We analytically build the connection between UAV jitter and its effects on mmWave channel;
- We propose a navigation-information-aided beam training for UAV mmWave communications;
- UAV beam training scheme assisted by navigation information can achieve better accuracy with reduced training length in AoA/AoD estimation.